Sina Yari

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Professional Summary

Ph.D. student and Graduate Research/Teaching Assistant in Computer Science at George Mason University. Research focuses on real-time and embedded systems, systems security, and multiprocessor scheduling. I design algorithms, mathematical models, and large-scale C/C++ & Python simulation frameworks for safety-critical systems, with current work mitigating schedule-based side-channel attacks through security-aware workload distribution across multiprocessor systems. Earlier research applied reinforcement-learning-based optimization for various performance objectives, including energy-, thermal-, and reliability-aware scheduling. Author of 9 peer-reviewed publications.

Research Interests

- Real-Time Embedded System Scheduling
- Operating System
- System Security
- Reliability-, Power-, and Thermal-Aware Optimization
- Machine Learning for System Optimization
- Computer System Architecture & HW/SW Co-Design
- Hardware-Level Debugging

Technical Skills

Programming Languages & Frameworks: C/C++, Java, Python, SQL, MATLAB, Rust, Verilog, VHDL, Bash, Embedded C, Linux Device Driver Development, JTAG, UART
Systems & Data: Git, MySQL, NumPy, Pandas
AI & Machine Learning: Reinforcement Learning, Genetic Algorithm
Simulation Tools: SLURM, CMake, Gem5, McPAT, HotSpot, Quartus, Xilinx, Design Compiler, Hspice, Pspice
Embedded/FPGA Platforms: Raspberry Pi, ARM Cortex, AVR, STM, NXP LPC, DE1, DE2, Spartan
Languages: Persian, English, German (Basic)

Education

George Mason University **Fairfax, VA**
Ph.D. in Computer Science (GPA: 3.81/4.0) *August 2022 – Present*
M.Sc. in Computer Science *2022 -2026*

Sharif University of Technology
M.Sc. Computer Engineering - Computer Systems Architecture (GPA: 4.0/4.0) *2018 - 2021*

Ferdowsi University of Mashhad
B.Sc. Computer Engineering - Hardware *2012 - 2017*

Work Experience

Graduate Research Assistant **August 2022 - Present**
George Mason University *Fairfax, VA*

- Developed large-scale C/C++ and Python discrete-event simulation frameworks for evaluating real-time scheduling, timing behavior, and system security.
- Designed and implemented security-aware scheduling techniques for real-time embedded systems to mitigate schedule-based side-channel attacks; published in **ISORC 2026** and **JSA 2023**.

Graduate Teaching Assistant **August 2022 - Present**
George Mason University *Fairfax, VA*

- Supported courses spanning Computer Systems Architecture, Advanced Computer Architecture, SWE for Real-Time Embedded Systems, Computer Communications and Networking, Database Systems, and Computer Systems Programming & Design (Java/Python/C). Held weekly recitations and office hours, providing feedback on code and assignments for 50+ undergraduate/graduate students.

Graduate Research Assistant **2018 - 2021**

Sharif University of Technology

- Designed energy-, thermal-, reliability-, and aging-aware scheduling techniques for real-time embedded systems to improve efficiency, safety, and fault tolerance.
- Developed and analyzed system-level optimization techniques, and explored reinforcement learning-based approaches for real-time embedded systems.

Graduate Teaching Assistant **2018 - 2021**

Sharif University of Technology

- Supported courses Fault-Tolerant Systems Design, Low Power Design, Digital Systems Design, Numerical Methods, Embedded Systems, Real-Time Systems, Logical Circuits Laboratory (as Instructor), and Digital Systems Design Laboratory (as Instructor). Held weekly office hours, providing feedback on code and assignments for 50+ undergraduate/graduate students.

Software Engineering Intern **May 2016 - August 2016**

Mini Computer Center (Peyman Nikkaran Tabaran Co.)

- Tested features for a zero client, mini computer, and thin client remote desktop application.
- Debugged and resolved connectivity/session handling issues affecting client deployments.

Undergraduate Research Assistant **2014 - 2017**

Ferdowsi University of Mashhad

- Analyzed the µC/OS-III task scheduler in C for an ARM Cortex-M3-based real-time embedded system.

Undergraduate Teaching Assistant **2014 - 2017**

Ferdowsi University of Mashhad

- Supported courses Logical Circuits, Machine and Assembly Languages, Computer Architecture. Held weekly office hours, providing feedback on code and assignments for 50+ undergraduate students.

Publications

- **S. Yari Karin**, H. Aydin, and D. Zhu, Task Splitting to Mitigate Schedule-Based Anterior Attacks in Real-Time Embedded Systems," *29th International Symposium on Real-Time Distributed Systems (ISORC)*, 2026.

- **S. Yari Karin**, H. Aydin, D. Zhu, S. Drager, and M. Anderson, Impact of Priority Assignment on Schedule-Based Attacks in Real-Time Embedded Systems," *Journal of Systems Architecture*, 2023.

- **S. Yari Karin**, R. Siyadatzadeh, M. Ansari, and A. Ejlali, Passive Primary/Backup-Based Scheduling for Simultaneous Power and Reliability Management on Heterogeneous Embedded Systems," *IEEE Trans on Sustainable Computing*, 2022.

- S. Safari, M. Ansari, H. Khdr, P. Gohari-Nazari, **S. Yari Karin**, A. Yeganeh-Khaksar, S. Hessabi, A. Ejlali, and J. Henkel, A Survey of Fault-Tolerance Techniques for Embedded Systems from the Perspective of Power, Energy, and Thermal Issues," *IEEE Access*, 2022.

- M. Ansari, S. Safari, **S. Yari Karin**, P. Gohari-Nazari, H. Khdr, M. Shafique, J. Henkel, and A. Ejlali, Thermal-Aware Standby-Sparing Technique on Heterogeneous Real-Time Embedded Systems," *IEEE Transactions on Emerging Topics in Computing*, 2021.

- M. Ansari, **S. Yari Karin**, S. Safari, and A. Ejlali, Power Management to Meet Thermal Safe Power in Fault-Tolerant Embedded Systems," *TechRxiv*, 2021.

- A. Yeganeh-Khaksar, M. Ansari, S. Safari, **S. Yari Karin**, and A. Ejlali, Ring-DVFS: Reliability-Aware Reinforcement Learning-Based DVFS for Real-Time Embedded Systems," *IEEE Embedded Systems Letters*, 2021.

- J. Saber-Latibari, M. Ansari, P. Gohari-Nazari, **S. Yari Karin**, A. M. H. Monazzah, and A. Ejlali, READY: Reliability- and Deadline-Aware Power-Budgeting for Heterogeneous Multicore Systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2021.

- **S. Yari Karin**, A. Sahraee, J. Saber-Latibari, M. Ansari, N. Rohbani, and A. Ejlali, "A Comparative Study of Joint Power and Reliability Management Techniques in Multicore Embedded Systems," *CSI/CPSSI International Symposium on Real-Time and Embedded Systems and Technologies (RTEST)*, 2020.

Awards & Honors

- **Student Travel Grant, ACM SIGMETRICS 2025** | *Poster presentation*, Stony Brook, New York, USA, 2025
- **Certificate in Inclusive Teaching for STEM Graduate Students** | *Completed certified training on inclusive and effective STEM teaching practices* (2025)
- **Top 0.1%, Nationwide M.Sc. Entrance Exam** | *Ranked 30th out of ≈ 30,000 in Computer Engineering Exam* (2018)
- **Top 5 M.Sc. Student, Computer Systems Architecture** | *Ranked 5th out of 27* (2021)